PRIMARY MATHEMATICS

Grade 4 Full Textbook (Chapters 1 - 4)

Chapter 1: Fractions and Place Value

1.1 Introduction to Fractions

A fraction represents a part of a whole. When an object or a group of items is divided into equal parts, each part is a fraction of the whole.

Core Definition:
• **Numerator:** The top number, showing how many equal parts are selected.
• **Denominator:** The bottom number, showing the total number of equal parts the whole is divided into.

Guided Example:
Express the fraction for a whole shaded block that has been sliced into 4 equal segments, where 3 segments are shaded.
Solution: The selected segments = 3 (Numerator). Total segments = 4 (Denominator). Written as: $\frac{3}{4}$

Practice Exercises:

1. Write down the numerator and denominator for $\frac{5}{8}$.
2. A farmer divided his circular shamba into 6 equal portions. He planted maize on 2 portions. What fraction of the shamba has maize?

1.2 Multi-Digit Place Value up to 10,000

Understanding large numbers requires looking at the position of each digit. Every position has a value ten times greater than the position to its right.

Ten Thousands (10,000s)	Thousands (1,000s)	Hundreds (100s)	Tens (10s)	Ones (1s)
4	3	8	2	5

Expanded Notation:
The number 43,825 can be written as:
40,000 + 3,000 + 800 + 20 + 5

Chapter 2: Multiplication and Division

2.1 Multi-Digit Multiplication (Area Model)

The Area Model helps us break down large numbers by their place values to make multiplication straightforward.

Guided Problem: Multiply 235 by 3 using the Area Model.
We expand 235 into **200 + 30 + 5**, then multiply each by 3:

×	**200**	**30**	**5**
3	600	90	15

Final Addition: 600 + 90 + 15 = **705**

Notebook Practice Tasks:

1. Calculate 142 × 4 using an area matrix box.
2. A market trader sells baskets for Sh. 315 each. If a customer buys 5 baskets, what is the total cost in shillings?

2.2 Introduction to Division with Remainders

When items cannot be shared perfectly evenly into equal whole numbers, the amount left over is called the **remainder**.

Key Concepts:

- **Dividend:** The total amount being divided.
- **Divisor:** The number you are dividing by.
- **Quotient:** The main whole number answer.
- **Remainder:** The leftover amount.

Chapter 3: Geometry and Measurement

3.1 Properties of 2D Shapes

Two-dimensional (2D) shapes are flat shapes that have length and width.

Core Properties:
- **Rectangle:** 4 sides, opposite sides are equal, and opposite sides are parallel.
- **Square:** 4 sides, all 4 sides are exactly equal in length.

3.2 Perimeter and Area

Perimeter is the total path boundary length around a shape. **Area** measures the surface space covered inside.

Measurement Breakdown:
For a rectangle with Length = 6 cm and Width = 4 cm:
- **Perimeter:** 6 + 4 + 6 + 4 = **20 cm**
- **Area:** 6 cm × 4 cm = **24 cm²**

Chapter 4: Decimals and Data Handling

4.1 Introduction to Decimals

Decimals are another way of writing fractional parts using a base-ten point tracking system.

Fraction-to-Decimal Mapping:

The mixed number $5\ \frac{6}{10}$ is written in decimal form as **5.6** (read as five point six).

4.2 Data Handling & Bar Graphs

Data handling involves gathering facts, organizing information, and presenting it visually to spot patterns clearly.

Tally Collection Practice:
Count the vegetable stock collected from the garden patch and draw corresponding block graphs:
• Tomatoes: |||| || (7)
• Potatoes: |||| |||| (9)